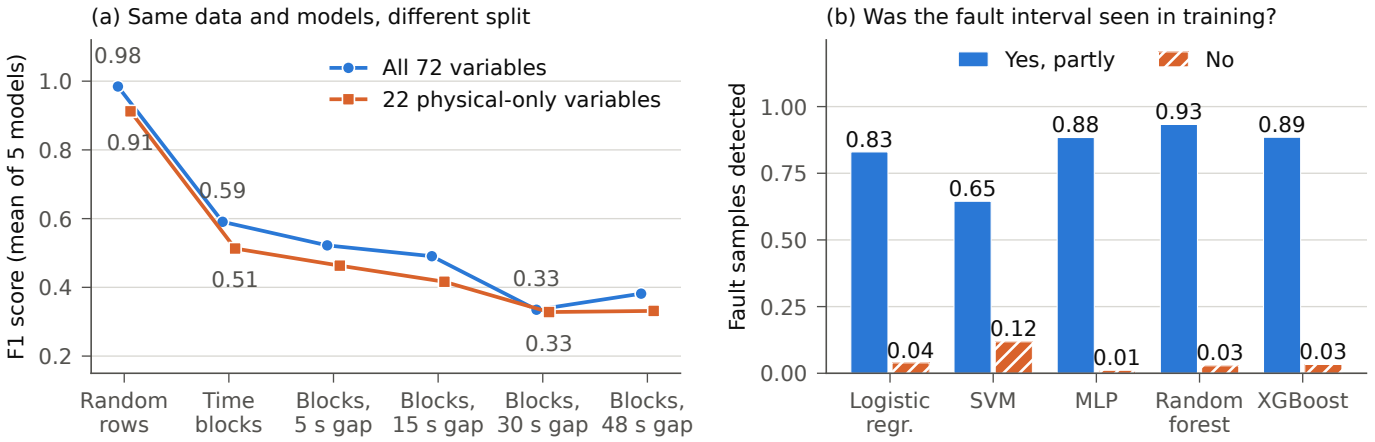
(a) Same data and models, different split
F1 score (mean of 5 models)
1.0
0.8
0.6
0.4
0.2
All 72 variables
22 physical-only variables
0.98
0.91
0.59
0.51
0.33
0.33
Random rows
Time blocks
Blocks, 5 s gap
Blocks, 15 s gap
Blocks, 30 s gap
Blocks, 48 s gap
(b) Was the fault interval seen in training?
Fault samples detected
Yes, partly
No
1.00
0.75
0.50
0.25
0.00
0.83
0.04
0.65
0.12
0.88
0.01
0.93
0.03
0.89
0.03
Logistic regr.
SVM
MLP
Random forest
XGBoost